

Windows machine. Duplicati is an open source backup client that provides an easy-to-use interface, supports strong encryption, and can send backups to various storage types including local folders, remote servers, and cloud providers. In this setup, Duplicati will be used to back up selected folders from the Windows system to the Ubuntu server using the SFTP protocol.

Visit the official website at <https://www.duplicati.com/download>, choose the Windows installer and download the .msi file. Run the installer and follow the default instructions to complete the installation.

After installation, Duplicati opens automatically in your default browser. If not, you can open it manually by visiting <http://localhost:8200>. This web interface is where all backup jobs are created, configured, and monitored. With Duplicati now installed and running, we can proceed to create the first backup job and link it to the Ubuntu backup server.

Creating the first backup job (with automation)

With Duplicati installed and the web interface open, the next step is to create a backup job that connects your Windows system to the Ubuntu backup server.

Initial setup: When Duplicati opens for the first time, it prompts you to secure the interface with a passphrase. While this step is optional if you plan to use Duplicati only from the same system, it is recommended for added protection. Click on 'Yes' to set a passphrase, then enter and confirm the password as instructed on screen. This ensures that only authorised users can access the Duplicati interface. Once the password setup is complete, the main dashboard appears with options on the left-hand side such as 'Home', 'Add backup', 'Restore', and 'Settings'.

Start a new backup job and fill general settings: To begin creating your backup job, click on *Add backup* from the left menu, choose *Configure a new backup* and click on *Next*. In this step, you define the basic details of your backup

job. Start by giving it a friendly name, such as 'SOHO Backup', and optionally add a description with any notes you find useful. For security, choose the built-in AES-256 encryption and set a strong passphrase to protect your backup data. This passphrase is required if you ever need to restore your files, so be sure to keep it safe and secure. Once all fields are filled, click *Next* to proceed.

Configure backup destination: In this step, you will specify where the backup will be stored. For this setup, the destination is the Ubuntu virtual machine configured earlier. Set the storage type to SFTP (SSH), then enter the IP address of the Ubuntu machine, such as 192.168.1.6. Use the default port 22 and provide the path on the server as */home/backupuser/duplicati-backups*. Enter the username as 'backupuser' and enter the corresponding password that was set for this account on the Ubuntu system. Click on *Test connection* to make sure Duplicati can reach the server. If successful, click on *Next* to proceed to the source folder selection.

Select source data: In this step, choose the folders on your Windows machine that you want to back up. Click on *Browse* and select important directories such as *Documents*, *Pictures*, or any

project-specific folders. You can include multiple folders as needed. It's best to avoid selecting system or temporary folders unless necessary. If you want to exclude certain files or file types, like *.tmp* or *.log*, you can click on *Edit Filters* and add the appropriate patterns; otherwise, you can leave the filters as they are. Once you've finalised your folder selections, click on *Next* to continue.

Set the backup schedule:

Automating backups: To ensure your backups run regularly without needing to start them manually, this step lets you schedule the process. Enable the option labelled 'Automatically run backups', set the frequency to every day, and choose a time when your computer is typically powered on, such as 2:00 AM. This ensures your data is consistently protected without requiring daily action from you.

You may adjust the frequency and time as needed based on your work pattern. Duplicati must be running (either as a background service or from the system tray) for scheduled tasks to execute. Click on *Next* to continue.

Configure options (optional): The final step includes advanced settings, most of which can be left at their default values. You may want to review a few options,

Figure 1: General and destination backup settings